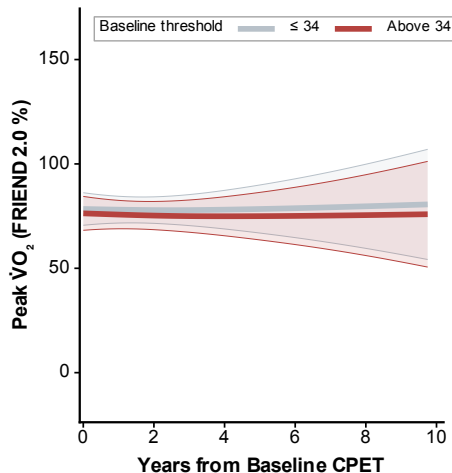


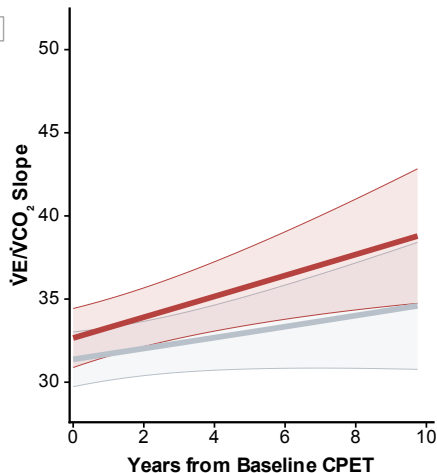
(A) Peak $\dot{V}O_2$

Baseline LAVi ≤ 34 mL/m² versus above 34 mL/m²



(B) $\dot{V}E/\dot{V}CO_2$ slope

Baseline LAVi ≤ 34 mL/m² versus above 34 mL/m²



(C) Longitudinal Model Summary

Model component	edf	F statistic	P value
Peak $\dot{V}O_2$			
Time smooth	1.18	0.02	0.945
Threshold group	—	0.39	0.534
Time x threshold group	1.01	0.02	0.918
$\dot{V}E/\dot{V}CO_2$ slope			
Time smooth	1.04	2.55	0.107
Threshold group	—	2.72	0.100
Time x threshold group	1.03	1.06	0.306

Supplemental Figure. Parameter-specific longitudinal cardiopulmonary trajectories stratified by binary baseline LAVi (mL/m²) status. **(A)** GAMM-derived Peak $\dot{V}O_2$ trajectories. **(B)** GAMM-derived $\dot{V}E/\dot{V}CO_2$ slope trajectories. **(C)** Summary of the key status and interaction terms for both models. Baseline threshold groups were defined as LAVi ≤ 34 mL/m² versus above 34 mL/m². Longitudinal models were adjusted for age, sex, and BMI.